OXIDACIÓN DE LOS ÁCIDOS GRASOS

- We are already aware of the importance of glucose as a fuel for tissues such as the brain, skeletal muscle, erythrocytes.

- However, for other tissues the oxidation of fatty acids to Acetyl CoA is a very important pathway.

- For example, the liver and heart obtain their energy mainly through the oxidation of fatty acids.

- The long chains of fatty acids are smaller than those of sugars, therefore, for the same weight, the oxidation of the carbons of a fatty acid generates more energy than those of glucose.

- 1 gram of fatty acid provides 9 Kcal; 1 gram of carbohydrate provides 5 kcal.

- The energy that is released in the oxidations of the carbons of a fatty acid to Ac.CoA is conserved in the form of NADH and $FADH_2$.

- As we already know, these cofactors subsequently give up the electrons to the electronic transport chain, which drives the synthesis of ATP.

- Likewise, the oxidation of Ac CoA in Krebs will drive the synthesis of more ATP.

- In some tissues, and in special circumstances, Ac.CoA can follow a different route from Krebs, and become ketone bodies.

- Ketone bodies are an alternative fuel that is exported to the brain and other tissues when there is no glucose

Fat digestion, mobilization and transport

- There are three sources of fatty acids to oxidize in cells:

    - dietary fats

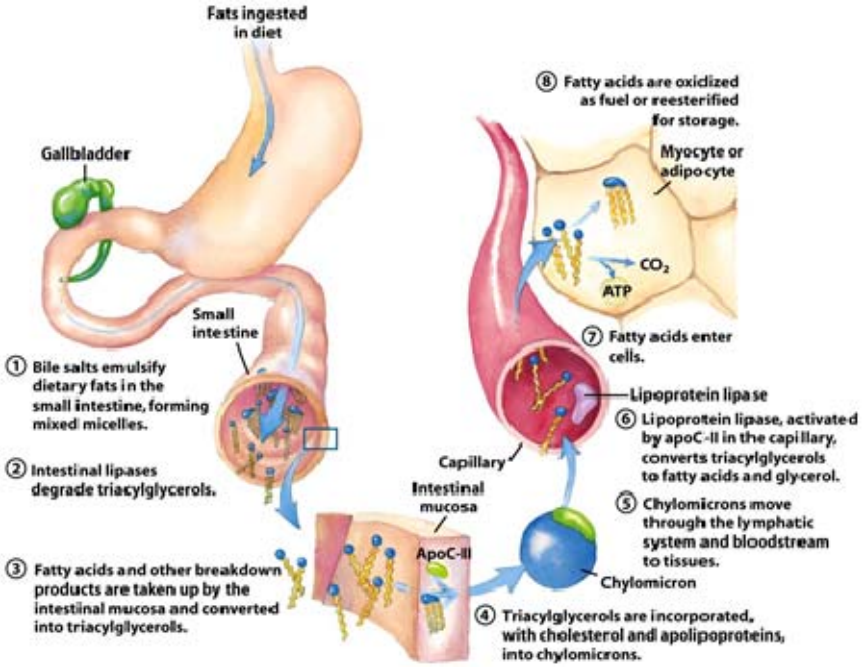
    - adipose tissue triacylglycerides

    - fats synthesized in the liver (mainly from excess glucose) that are exported to other tissues

Fat absorption from the diet

(You have seen this section with Dr Pío. It is a mere reminder)

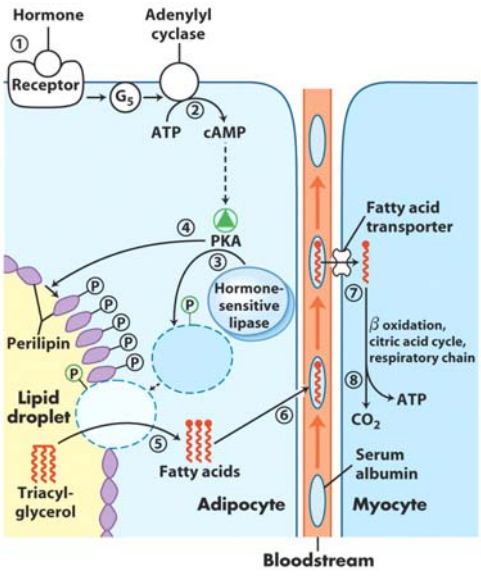
- Due to their insolubility, fats reach the intestine intact and, in order to be absorbed by the intestinal epithelium, they must be solubilized by bile salts.
- The amphipathic character of bile salts causes the fats to form micelles, which increases their accessibility to intestinal lipases.
- The action of lipases converts TAGs sequentially into DAG, MAG and, finally, into free fatty acids and glycerol.



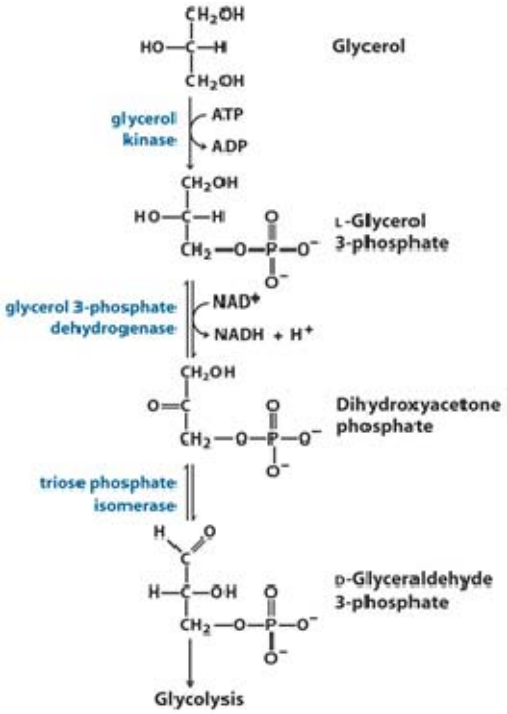
- Then fatty acids and glycerol diffuse into the epithelial cell where they are reconverted into TAG.
- Finally, TAGs, together with dietary cholesterol and specific proteins, form lipoprotein aggregates called chylomicrons.
- Chylomicrons transport TAGs and cholesterol to the liver, adipose tissue, etc. where, through the action of some of the apoproteins that are part of the chylomicron (apoC-II), they activate lipoprotein lipases that hydrolyze the TAGs again to fatty acids and glycerol, which are finally absorbed by the cells of the target tissues.
- The subsequent fate of fatty acids depends on the tissue. In the case of adipose, a new re-esterification to TAG occurs and they are stored; in the case of muscle, the fatty acids are oxidized.

Mobilization of stored fats

- When hormones signal the existence of a need for energy, the body responds by mobilizing the TAGs that are stored in adipose tissue, to provide fatty acids to the muscle, heart, etc.
- Hormones such as adrenaline and glucagon, which are released in hypoglycemic situations, activate adipocyte adenylate cyclase, which leads to an intracellular increase in cAMP.

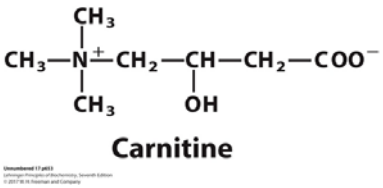


- The cAMP will activate the PKA which will finally cause the phosphorylation of a protein present in the lipidic droplet called perilipin.
- Phosphorylated perilipin attracts cytosolic lipase to TAGs, which hydrolyzes TAGs to fatty acids and glycerol.
- Fatty acids go out into the bloodstream where they bind to serum albumin that moves them to peripheral tissues such as liver, muscle, heart and adrenal cortex where they are oxidized.
- Glycerol can also provide energy through its oxidation **by glycolysis**

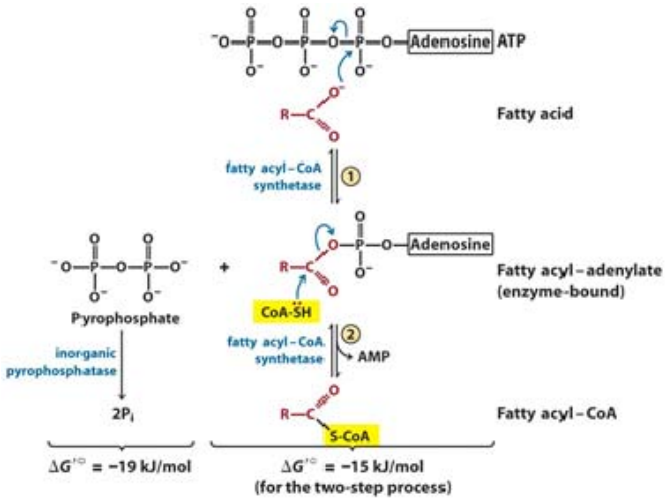


Activation and mobilization of fatty acids for their oxidation

- The enzymes necessary to oxidize fatty acids are located in the mitochondrial matrix (Kennedy and Lehninger 1948).
- Fatty acids with 12 or less carbons can enter the mitochondria without an specific transport system.
- On the other hand, fatty acids with 14 or more carbons cannot pass the inner mitochondrial membrane and must be transported by a system known as the **carnitine shuttle.**
- Carnitine is an organic molecule (4-trimethyl amino, 3 hydroxy butyrate) that binds to fatty acids on the cytosolic side and transports them inside the mitochondrial matrix.

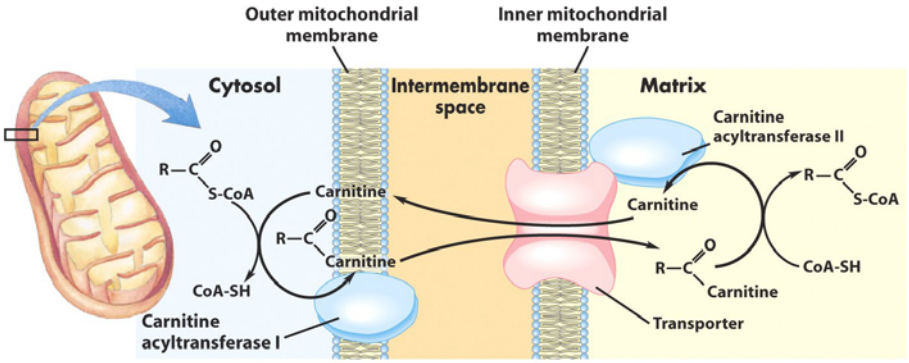


- First, the fatty acid is activated by its binding to CoA.
- This reaction is catalyzed by a family of isoenzymes present in the outer mitochondrial membrane, called **fatty acyl-CoA synthetases**
- These enzymes catalyze the formation of a thioester between the C carboxylic acid. fatty acid and the -SH of CoA to generate a fatty acyl-CoA.
- In the reaction, one molecule of ATP is completely hydrolyzed to AMP and PPi, **which implies the net expenditure of two ATPs**. (It will be necessary to use one ATP molecule to phosphorylate AMP to ADP. Then ADP is phosphorylated to ATP by the oxidative phosphorylation)



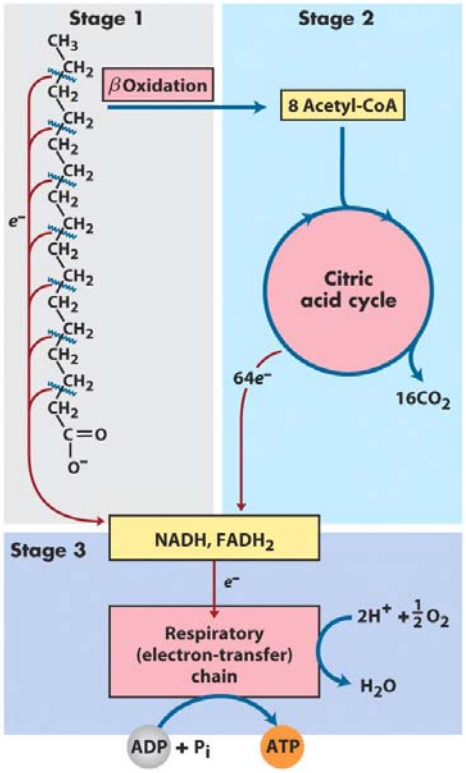
- The formation of this thioester bond causes the synthesis of a high energy compound (fatty acyl-CoA)
- Subsequently, the fatty acyl-CoA is transferred to the mitochondrial matrix through the carnitine-acylcarnitine transporters through a process that involves two transesterifications of the fatty acid
- First carnitine acyl transferase I (CATI), on the outer face of the inner mitochondrial membrane, transfers the fatty acid from the CoA to the OH of carnitine, forming acyl-carnitine.
- - Second, acyl-carnitine enters the matrix through the carnitine-acylcarnitine transporter at the inner mitochondrial membrane
- Finally, in the matrix, the fatty acid is transferred from carnitine to a new CoA molecule. This reaction is catalyzed by carnitine acyl transferase II (CATII), located on the inner face of the inner mitochondrial membrane.

- **This transport of the fatty acid from the cytoplasm to the matrix is the stage that limits the rate of oxidation of fatty acids** in the mitochondria. Once the fatty acid is in the matrix, its only fate is oxidation.
- **The principal allosteric regulator of CATI is malonyl-CoA** , which is the molecule used to synthesize fatty acids in the cytoplasm. This fact is responsible for the coordinated regulation of the synthesis and oxidation of fatty acids



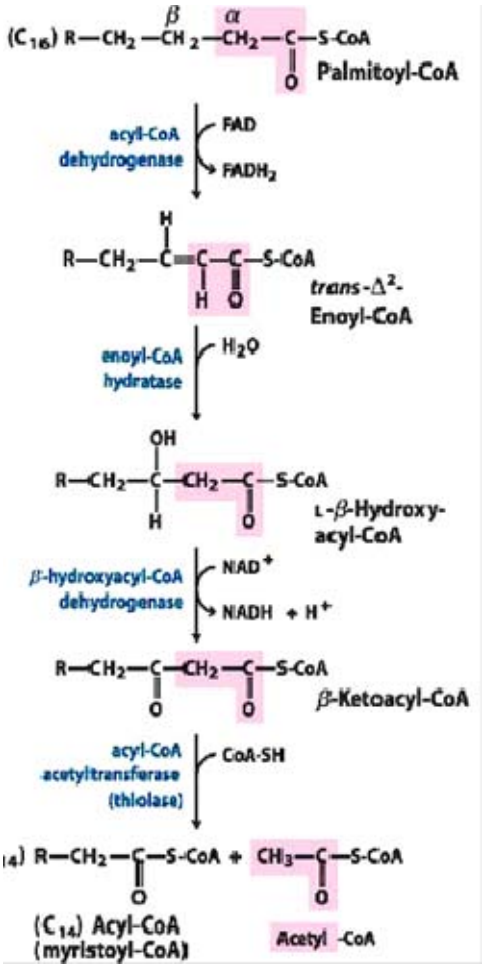
- acids.

Fatty acid oxidation (β-oxidation)



- The complete oxidation, to CO_2, of fatty acids carbons occurs in three stages:
- - **In the first**, **called β-oxidation**, two-carbon units are successively eliminated in the form of acetyl-CoA, starting at the carboxyl end of the fatty acid. E.g. palmitic (16C) undergoes 7 cycles of β-oxidation and yields 8 molecules of acetyl-CoA.
- **In the second phase**, the acetyl groups of acetyl-CoA are oxidized to CO_2 through the **Krebs cycle**.
- **In the third phase**, all the NADH and $FADH_2$ molecules produced during the oxidative reactions of the two previous phases, channel the electrons to the respiratory chain, and allows the synthesis of ATP at the oxidative phosphorylation process.

β-oxidation of saturated fatty acids



- Each cycle of β-oxidation consists on four reactions:
- Firstly, the C2 and C3 of the fatty acid are oxidized by the action **of acyl-CoA dehydrogenase.** In this reaction, a trans unsaturated isomer called trans-Δ^2-enoyl-CoA is generated.
    - o They're three isozymes of acyl-CoA dehydrogenase, each with specificity for fatty acids with an length range:
        - VLCAD: 12-18 C
        - MCAD: 4-14 C
        - SCAD: 4-8 C
    - o the three isozymes have FAD as an oxidation-reduction cofactor and the electrons released from the fatty acid immediately pass to the electron transport chain through a protein called electron transfer flavoprotein (ETF).Therefore, these electrons will generate 1.5 ATP in oxidative phosphorylation

- **In the second reaction**, a molecule of H_2O is added stereospecifically to the double bond to form the L-stereoisomer of **β-hydroxyacyl-CoA**. This reaction is catalyzed **by Enoyl-CoA hydratase**.
- **The third reaction consists on the dehydrogenation** (oxidation) of carbon 3 **(also called β-carbon)** of β-hydroxyacyl-CoA **to form β-ketoacyl-CoA**. This reaction is catalyzed by **hydroxyacyl-CoA dehydrogenase. In this case the electron acceptor is NAD^+**.
- **The fourth and last reaction is catalyzed by acyl-CoA acetyltransferase (or thiolase)** and cleaves the bond between carbons 2 and 3 through a reaction in which the -SH of the CoA forms a thioester with the carbonyl carbon forming Ac.CoA and a fatty acylCoA with two less carbons.
- In summary, **the three initial reactions of each β-oxidation turn transforms a $–CH_2-$ group into a carbonyl group**, which makes the C2-C3 bond much less stable, and susceptible to the nucleophilic attack of the CoA –SH of that takes place in the reaction catalyzed by thiolase
- The first three reactions are analogous to those catalyzed by succinate dehydrogenase, fumarate hydratase and malate dehydrogenase at the Krebs cycle
- The β-oxidation cycles continue until all the carbons of the fatty acid remain as Ac.CoA.
- The balance for the oxidation of Palmitil-CoA up to eight ac. CoA will be:

$Palmitoyl\text{-}CoA+7CoA+7FAD+7NAD^+ +7H_2O \rightarrow 8\ ac.CoA+ 7FADH_2+7NADH+H^+$

- Additonally, the Ac.CoA can continue to oxidize in the Krebs cycle, so the total yield for oxidation will be 108 ATP

TABLE 17–1 Yield of ATP during Oxidation of One Molecule of Palmitoyl-CoA to CO_2 and H_2O

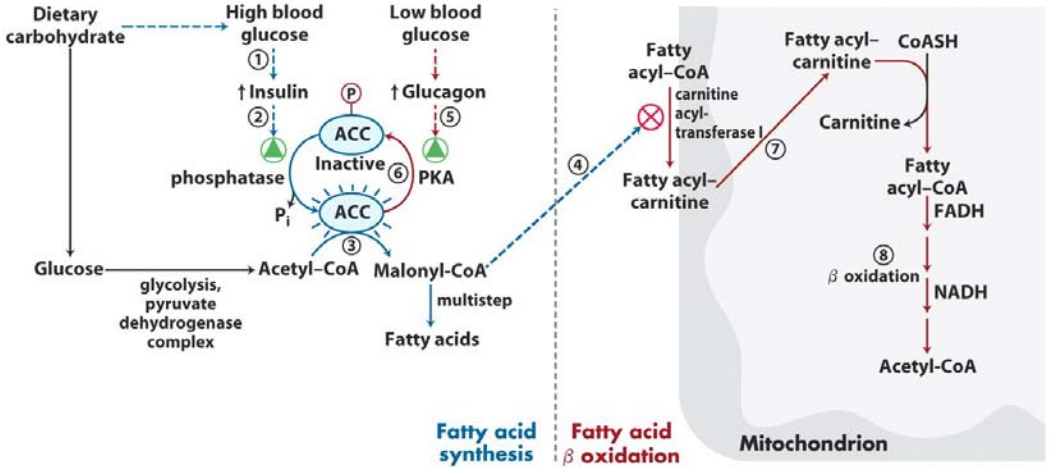
Enzyme catalyzing the oxidation step	*Number of NADH or $FADH_2$ formed*	*Number of ATP ultimately formed**
Acyl-CoA dehydrogenase	7 $FADH_2$	10.5
β-Hydroxyacyl-CoA dehydrogenase	7 NADH	17.5
Isocitrate dehydrogenase	8 NADH	20
α-Ketoglutarate dehydrogenase	8 NADH	20
Succinyl-CoA synthetase		8†
Succinate dehydrogenase	8 $FADH_2$	12
Malate dehydrogenase	8 NADH	20
Total		108

A

- However, we must consider that in the cytoplasmic activation of palmitate two ATP will be spent. Thus, the net yield is 106 ATP

Regulation of fatty acid oxidation

- In the liver, the acyl-CoA originated in the cytosol during the biosynthesis of fatty acids can follow two routes:
  - β-oxidation in the mitochondria
  - conversion to TAG and phospholipids in the cytoplasm
- The route followed will depend on the speed of long-chain fatty acids to the mitochondria via the carnitine shuttle



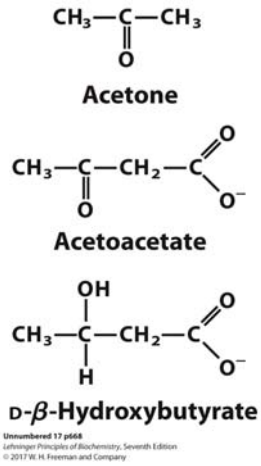
- CATI is responsible for the transport of fatty acid from the cytoplasm to the mitochondria. This enzyme is regulated by the levels of malonyl-CoA synthesized by the enzyme called Acetyl CoA carboxylase involved in the synthesis of fatty acids.
- When blood glucose levels are low (hypoglycemia), glucagon will cause an intracellular cascade that, by raising cAMP, will activate PKA and this inhibits the synthesis of Malonil CoA in the cytoplasm. In these circumstances, CATI will work at full capacity and will facilitate the entry of fatty acids into the mitochondria for oxidation. We will come back to this during the fatty acid synthesis

- Pathologies caused by Fatty acyl-CoA dehydrogenases defects
- One of the most frequent genetic defects that affect the catabolism of fatty acids is the mutation in the gene that encodes for the medium chain acyl-CoA dehydrogenase (MCAD).
- This mutation generates fat accumulation in the liver, low blood glucose, vomiting and coma.
- There are cases of mutations in other enzymes of this pathway, which produce a deficit in the ability to obtain energy from fat oxidation. This type of deficiency causes serious heart disease and skeletal muscle disorders.

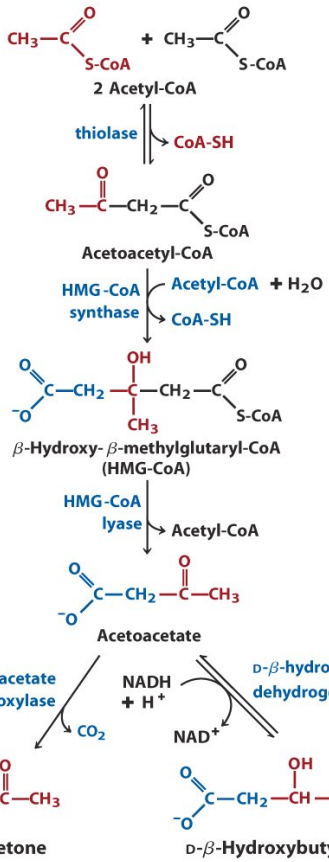
- Fat oxidation in peroxisomes
- Although the oxidation of fatty acids take place mainly in the mitochondria, there are other organelles such as peroxisomes where they can also be oxidized, especially the long-chain ones.
- Peroxisomal β-oxidation reactions are identical to those of mitochondrial but there are some differences in the mechanisms
- In the first reaction, the FADH2 the electrons reduce directly the O_2 and produces H_2O_2. This compound is dangerous and must be removed immediately by catalase.
- In mammals, peroxisomal enzymes are active on very long chain fatty acids. Its function is to shorten the length of these fatty acids.
- Since peroxisomes lack Krebs enzymes, once their length is shortened, the fatty acids are transferred to the mitochondria to be completely oxidized.
- In these organelles there is also the oxidation of ac. phytanic, a branched fatty acid very abundant in dairy products, meat and fish.
- The oxidation of phytanic is called α-oxidation.
- There are several pathologies caused by defects in the metabolism of peroxisomes:
    - **Zellweger syndrome**: peroxisomes cannot be made
    - **X-linked adrenoleukodystrophy**: peroxisomes cannot oxidize long-chain fatty acids
    - **Refsum's disease**: cannot metabolize phytanic acid

Ketone bodies

- During periods of high activity the Ac.CoA produced by the β-oxidation is fundamentally oxidized in Krebs in

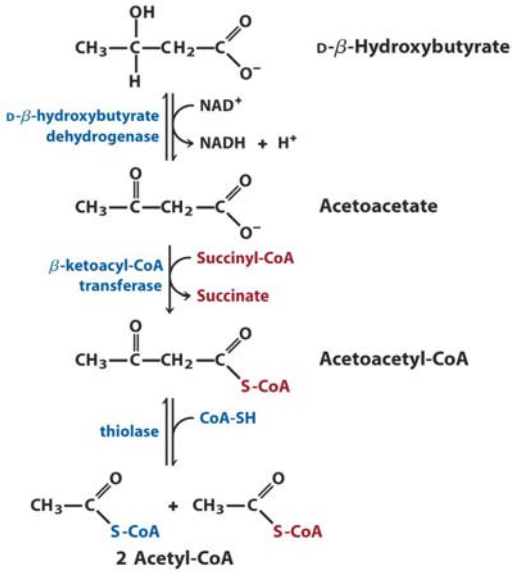


- But, in the liver, there is an alternative fate whereby the Ac.CoA molecules can also become ketone bodies: **acetone, acetoacetate and D-β-hydroxybutyrate**.
- The final destination of ketone bodies is to be exported to other tissues such as skeletal muscle, cardiac, and adrenal cortex where they are converted into Ac-CoA and oxidized in Krebs.
- The brain, when there is not enough glucose, can be adapted to the use of acetoacetate and hydroxybutyrate for a short period of time
- The synthesis of ketone bodies occurs in the liver. **The first step consists of the condensation of two molecules of Ac. CoA**, through the action of thiolase, to give rise to acetoacetyl-CoA.



- **Afterwards, acetoacetyl-CoA is condensed with a third molecule of Ac.CoA** to form **β-hydroxy- β-methylglutaryl-CoA**, by **HMG-CoA synthase**
- **Then the HMG-CoA breaks down giving rise to acetoacetate and Ac.-CoA**.
- **Acetoacetate can be reduced to D-β-hydroxybutyrate by D-β-hydroxybutyrate dehydrogenase.** A part of the acetoacetate can be decarboxylated **to acetone**, which is highly volatile and is exhaled in the lungs.
- **In healthy people the amount of acetone produced is low**. **In diabetics, a large amount of acetoacetate is produced and, therefore, acetone.** This is why their breath smells of this compound

- - Both acetoacetate and hydroxybutyrate are soluble so, via blood, they reach peripheral tissues where acetoacetate is activated forming an ester with CoA, and generating acetoacetylcoA through β-ketoacyl-CoA transferase
- Finally, thiolase, cleaves the acetoacetyl-CoA producing two acetyl-CoA molecules that are oxidized in Krebs



- This route is especially intense when, as a consequence of very continuous hypoglycemia, the liver must synthesize glucose from oxaloacetate.
- The exit of oxaloacetate from the Krebs cycle leads to a limited possibility to oxide Ac CoA.
- The conversion of Ac.CoA into ketone bodies makes it possible, on the one hand, to provide alternative fuels to the peripheral tissues, and, on the other, to maintain the oxidation of fatty acids in the liver.
- This process is especially intense in situations such as untreated diabetes. In these situations, the excessive increase of ketone bodies synthesis causes a decrease of blood pH, causing acidosis that can lead to coma.
- Diets low in carbohydrates can also lead to ketoacidosis.

